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Electronic Companion to "When Front-Stage Innovation Overloads Back-Stage Workers"

(Authors' names are not included for peer review)

Abstract. This electronic companion provides full proofs, robustness analyses, and calibration details for the main paper.

Key words: electronic companion

1. Full Proofs

We restate each result for convenience and provide complete proofs. The model equations are:

$$D = (1 + aT) \cdot F^b, \tag{1}$$

$$G = D - K \cdot h(W), \quad h(W) = (1-p) + pW, \tag{2}$$

$$\frac{dK}{dt} = r\,[G]^+\,W - \delta K + u, \tag{3}$$

$$\frac{dW}{dt} = ([-G]^+ + f)(1-W) - (\mu + \sigma\,[G]^+)\,W, \tag{4}$$

$$\frac{dT}{dt} = \beta\,[-G]^+\,(1-T) - \eta\beta\,[G]^+\,T - \lambda T. \tag{5}$$

Customer lifetime value is denoted CLV throughout; starred variables (K^*, W^*, T^*) denote steady-state equilibrium values.

1.1. State-Space Invariance

LEMMA 1 **(Forward Invariance).** *The set* $\Omega = \{(K,W,T) : K > 0,\ W \in [0,1],\ T \in [0,1]\}$ *is forward-invariant under the flow of the ordinary differential equation (ODE) system* (3)–(5) *for all* $t \geq 0$*, provided* $u > 0$*,* $f \geq 0$*, and all parameters are non-negative.*

A t each boundary, the vector field points into Ω: $\dot{W}|_{W=0} = [-G]^+ + f \geq 0$; $\dot{W}|_{W=1} = -(\mu + \sigma[G]^+) \leq 0$; $\dot{T}|_{T=0} = \beta[-G]^+ \geq 0$; $\dot{T}|_{T=1} = -(\eta\beta[G]^+ + \lambda) \leq 0$; $\dot{K}|_{K=0} = r[G]^+W + u > 0$. Forward invariance follows by the standard barrier argument. □

REMARK 1 (PIECEWISE-SMOOTH DYNAMICS). The system (3)–(5) is piecewise-smooth, with a non-smooth gradient at the switching manifold $\mathcal{M} = \{G = 0\}$. Existence and uniqueness of solutions follow from Filippov's (1988) framework, provided trajectories cross $\mathcal{M}$ transversally; numerical verification at both equilibria confirms this under baseline parameters. The implicit function theorem arguments underpinning the analytical results are applied at surplus equilibria where $G^* = -S^* < 0$, so differentiability is guaranteed by the smooth structure of the ODE within $\{G < 0\}$; smoothing robustness is verified in §1.8.

REMARK 2 (COMPUTABILITY OF F_{crit}). The implicit balance condition $K^*h(W^*) = (1 + aT^*)F^b + S^*$ rearranges to $F = (K^*h(W^*)/(1 + aT^*))^{1/b}$, but this is self-referential since W^* and T^* depend on F.

The computable form evaluates h and the demand term at the boundary $S^* \to 0$ where $T^* \to 0$, yielding the closed-form threshold $F_{\text{crit}} = (K^* h(f/(f+\mu)))^{1/b}$.

1.2. Equilibrium Existence and Uniqueness

Before proving the main theorems, we establish that the surplus equilibrium invoked by Theorems 1–3 exists and is unique. The equilibrium expressions derived in §4.1 of the main paper are conditional on a surplus $S^* > 0$, but S^* itself depends on the equilibrium values W^* and T^* through $S^* = K^* h(W^*) - D(T^*, F)$, creating a fixed-point problem. We resolve this by reducing the system to a single implicit equation in S^* and establishing that it has exactly one positive solution.

LEMMA 2 (**Surplus Equilibrium Existence; Uniqueness under C2**). *For any $F < F_{\text{crit}}$, a surplus equilibrium (K^*, W^*, T^*) with $S^* > 0$ exists. Under Regularity Condition C2 (single-crossing condition C2-SC: $\beta(f+\mu) < \lambda$; analytic, see Step 1 of the Theorem 1 proof, EC §EC.1.3), this equilibrium is unique.*

A t surplus equilibrium, $K^* = u/\delta$. The equilibrium conditions yield $W^*(S) = (S+f)/(S+f+\mu)$ and $T^*(S) = \beta S/(\beta S+\lambda)$, both continuous and strictly increasing in surplus S. Define the surplus residual $\Phi(S) = (u/\delta)[(1-p) + pW^*(S)] - (1+aT^*(S))F^b - S$. Then $\Phi(0) > 0$ when $F < F_{\text{crit}}$, and $\Phi(S) \to -\infty$ as $S \to \infty$. Existence follows by IVT. For uniqueness, $\Phi'(S) = \tau_1(S) - \tau_2(S) - 1$ where $\tau_1, \tau_2 > 0$ are decreasing. Under C2-SC ($\beta(f+\mu) < \lambda$), the equation $\Phi'(S) = 0$ has at most one solution (a decreasing function equals an increasing function), so Φ has exactly one positive root. C2-SC holds for all $f \in [0, 0.10]$: $\beta(f+\mu) \leq 0.018 < 0.025 = \lambda$. $\square$

REMARK 3 (LOW-TRUST INTERIOR ATTRACTOR). A second fixed point exists for large F, with $T^*{=}0$ and $W^*_{\text{low}} = f/(f+\mu+\sigma G^*)$. The Jacobian has strictly negative eigenvalues (verified numerically for $F \in [1.0, 1.5]$). This interior attractor, jointly with the surplus equilibrium of Lemma 2, establishes the bistability invoked by Theorem 3.

LEMMA 3 (**Forward-Boundedness of Trajectories**). *For any finite F and any initial condition $(K_0, W_0, T_0) \in \mathbb{R}_{>0} \times [0,1] \times [0,1]$, the solution satisfies $K(t) \leq \max(K_0, C(F)/\delta)$ for all $t \geq 0$, where $C(F) = r(1+a)F^b + u$. Combined with $W(t) \in [0,1]$ and $T(t) \in [0,1]$ (both preserved by the ODE structure), the state space is forward-invariant and bounded.*

U nder overload ($G > 0$), we bound $G^+ = D(T,F) - K\,h(W) \le D(T,F) \le (1+a)F^b$ (since $T \le 1$) and $W \le 1$. Therefore

$$\begin{aligned}\dot{K} &= rG^+W - \delta K + u \\ &\le r(1+a)F^b - \delta K + u = C(F) - \delta K.\end{aligned}$$

By Gronwall's lemma, $K(t) \le \max(K_0,\, C(F)/\delta)$ for all $t \ge 0$. Under surplus ($G \le 0$), $G^+ = 0$ and $\dot{K} = -\delta K + u \le 0$ whenever $K \ge u/\delta$, so K is bounded above by $\max(K_0, u/\delta)$. In both regimes the upper bound $\max(K_0, C(F)/\delta)$ holds. At baseline parameters with $F = 1.2$, this gives $K_{\max} \approx 25.3$, far above the operating range $K^* = 1.67$. Since W and T are bounded to $[0,1]$ by construction, the full state space is forward-invariant and compact for all finite F. $\square$

REMARK 4 (SCOPE OF GLOBAL CONVERGENCE). Lemma 3 ensures all trajectories remain in a bounded compact set. Which attractor a trajectory reaches depends on its initial condition relative to the basin boundary (Figure 2, main paper). For $F \ge F_{\text{crit}}$, only the low-trust attractor is stable (Remark 3). A complete global Lyapunov proof is left to future work; the principal obstacle is the switching manifold $\{G{=}0\}$ where the standard LaSalle argument does not apply directly, requiring Filippov-specific Lyapunov construction (Filippov 1988).

REMARK 5 (MAINTAINED CONDITIONS). Four conditions are maintained throughout: (C1) $\Phi(0) > 0$ (i.e., $F < F_{\text{crit}}$; analytic via IVT); (C2) uniqueness of S^*, established analytically under C2-SC ($\beta(f+\mu) < \lambda$, which holds for all $f \in [0, 0.10]$ independent of b, p, σ; see Step 1 of the Theorem 1 proof, EC §EC.1.3); (C3) regularity $\Phi'(S^*) < 0$, a consequence of (C1)–(C2), required for the implicit function theorem in Theorems 1–2; and (C4) low-trust attractor existence $F^b > (u/\delta)\,h(W^*_{\text{low}})$, verified numerically (Jacobian eigenvalues across $F \in [1.0, 1.5]$).

REMARK 6 (BIFURCATION CLASSIFICATION AT F_{crit}). The transition at F_{crit} is a saddle-node (fold) bifurcation in the surplus variable S: the stable surplus equilibrium and the separating saddle coalesce as $F \nearrow F_{\text{crit}}$. The (W,T) sub-system eigenvalues retain strictly negative real parts throughout $F \in [0, F_{\text{crit}}]$ (baseline: complex-conjugate pair with $\text{Re} \approx -0.07$; characteristic polynomial constant term > 0 up to F_{crit}), ruling out Hopf bifurcation.

1.3. Proof of Theorem 1 (Front-Stage Trap)

THEOREM 1 **(Front-Stage Trap, restated).** *Suppose $b > 1$ and $p > 0$. Define the threshold $F_{\text{crit}} = \big((u/\delta)\,h(f/(f+\mu))\big)^{1/b}$. Then: (i) for $F < F_{\text{crit}}$, the unique surplus equilibrium exists with $T^* > 0$ and $\text{CLV}^* > \text{CLV}_{\text{low}}$; (ii) for $F \geq F_{\text{crit}}$, no surplus equilibrium exists; all forward trajectories remain bounded (Lemma 3) and converge numerically to the low-trust attractor with $\text{CLV} \approx \text{CLV}_{\text{low}}$.*

W e show that the surplus equilibrium exists if and only if $F < F_{\text{crit}}$, and that beyond this threshold the system is attracted to the low-trust fixed point.

Step 1: Surplus equilibrium requires $\Phi(0) > 0$. From Lemma 2, the surplus equilibrium exists if and only if $\Phi(0) = (u/\delta)\,h(f/(f+\mu)) - F^b > 0$, which rearranges to $F < F_{\text{crit}} = \big((u/\delta)\,h(f/(f+\mu))\big)^{1/b}$. (Here we use the fact that at $S^* = 0$, $T^* \to 0$ and $W^* \to f/(f+\mu)$.)

Step 2: Within the surplus regime, surplus decreases in F. At the surplus equilibrium, $K^* = u/\delta$ is independent of F. Differentiating the surplus residual, $dS^*/dF < 0$ for F near F_{crit} because the demand derivative $b(1+aT^*)F^{b-1}$ dominates the indirect effects through W^* and T^*. As $F \to F_{\text{crit}}$, $S^* \to 0$.

Step 3: Beyond F_{crit}, overload drives collapse. For $F \geq F_{\text{crit}}$, $G > 0$, and from (5):

$$\frac{dT}{dt} = -T(\eta\beta G + \lambda) < 0 \quad \text{for all } T > 0.$$

Trust declines monotonically. Through Loop R2 ($W\!\downarrow\to h(W)\!\downarrow\to G\!\uparrow$), the gap widens, accelerating trust decline. The convexity of demand in F ($\partial^2 D/\partial F^2 > 0$ for $b > 1$) ensures the gap widens super-linearly, making the collapse self-reinforcing under Loop R2; convergence to the low-trust attractor is confirmed numerically across the studied parameter configurations (Tier C; main paper Remark 1). The system converges to the low-trust interior attractor (Remark 3).

Boundary conditions. When $b \leq 1$: demand is concave in F; the balancing loop B1 can compensate and the regime shift does not occur. When $p = 0$: $h(W) = 1$, severing Loop R2; the system may still experience trust decline through the direct overload channel but without the workforce-mediated amplification.

1.4. Proof of Theorem 2 (Good Jobs Buffer)

THEOREM 2 **(Good Jobs Buffer, restated).** *Suppose $p > 0$ and the surplus equilibrium is unique and stable (Regularity Condition: $\Phi'(S^*) < 0$; Lemma 2). Then $\partial\,\text{CLV}^*/\partial f > 0$ for all $f \geq 0$.*

W e show that each link in the chain $f \to W^* \to h(W^*) \to K_{\text{eff}} \to S^* \to T^* \to \text{CLV}^*$ is strictly positive.

Link 1: $\partial W^*/\partial f > 0$**.** At the surplus equilibrium, from (4) with $[G]^+ = 0$:

$$\begin{aligned} 0 &= S^*(1-W^*) + f(1-W^*) - \mu W^* \\ &= (S^*+f)(1-W^*) - \mu W^*. \end{aligned}$$

Solving: $W^* = (S^*+f)/(S^*+f+\mu)$. Differentiating: $\partial W^*/\partial f = \mu/(S^*+f+\mu)^2 > 0$, positive and decreasing in f (diminishing returns).

Link 2: $\partial h/\partial W = p > 0$**.** By construction of $h(W) = (1-p) + pW$.

Link 3: $\partial S^*/\partial h = K^* > 0$**.** Since $S^* = K^* h(W^*) - D(T^*, F)$ and $K^* = u/\delta > 0$.

Link 4: $\partial T^*/\partial S^* > 0$**.** From the equilibrium expression $T^* = \beta S^*/(\beta S^* + \lambda)$: $\partial T^*/\partial S^* = \beta\lambda/(\beta S^* + \lambda)^2 > 0$.

Link 5: $\partial\,\text{CLV}/\partial T^* > 0$**.** By assumption on the CLV functional form (w and ρ both increasing in T).

Combining links via the chain rule:

$$\frac{\partial\,\text{CLV}^*}{\partial f} = \underbrace{\frac{\partial\,\text{CLV}}{\partial T^*}}_{>0} \cdot \underbrace{\frac{\partial T^*}{\partial S^*}}_{>0} \cdot \underbrace{K^*}_{>0} \cdot \underbrace{p}_{>0} \cdot \underbrace{\frac{\partial W^*}{\partial f}}_{>0} > 0.$$

Diminishing returns: $\partial^2\,\text{CLV}^*/\partial f^2 < 0$ follows from the concavity of W^* in f (the $(S^*+f+\mu)^{-2}$ factor is decreasing in f) propagated through the chain.

General-equilibrium correction. The full equilibrium embeds a feedback loop $f \to W^* \to h \to S^* \to T^* \to D$, which partially offsets the direct surplus gain by raising demand through higher trust. To establish that this feedback does not reverse the sign, we apply the implicit function theorem directly.

Sign established analytically. The equilibrium condition $\Phi(S^*) = 0$ defines S^* implicitly as a function of f. Differentiating:

$$\frac{dS^*}{df} = -\frac{\partial\Phi/\partial f}{\Phi'(S^*)}.$$

Since $\partial\Phi/\partial f = (u/\delta)\, p\, \mu/(S^*+f+\mu)^2 > 0$ and $\Phi'(S^*) < 0$ (Regularity Condition C3), we have $dS^*/df > 0$ for all $f \geq 0$. The sign of $d\,\text{CLV}^*/df$ is therefore analytically established; it does not depend on the magnitude of the trust-demand feedback.

Magnitude of the GE correction (numerically confirmed). The trust-demand feedback damps the surplus gain by a factor proportional to $aF^b\,\beta\lambda/(\beta S^* + \lambda)^2/|\Phi'(S^*)|$, which we refer to as the GE correction ratio. At baseline parameters this ratio is approximately 0.09, confirmed across 100 parameter configurations spanning the empirically relevant range. The correction dampens but does not reverse the direct effect; $d\,\mathrm{CLV}^*/df > 0$ throughout.

1.5. Proof of Theorem 3 (Burnout Cliff)

THEOREM 3 **(Burnout Cliff, restated).** *Suppose $\sigma > 1$, $p > 0$, and $F_0 < F_{\text{crit}}$ (a surplus equilibrium exists at F_0). Consider a firm at the surplus equilibrium that implements a step increase in F from F_0 to $F_0 + \Delta F$. The outcome is governed by two thresholds:*

(i) Safety threshold (permanent-collapse onset). *For*

$$\Delta F \;\geq\; \Delta F_{\text{collapse}} = F_{\text{crit}}(f) - F_0$$
$$= \Big(\frac{u}{\delta}\,h\Big(\frac{f}{f+\mu}\Big)\Big)^{1/b} - F_0,$$

no surplus equilibrium exists at the post-upgrade level; the low-trust attractor is the only locally stable state (Remark 3), and convergence to it is confirmed numerically across the studied initial conditions. For $\Delta F < \Delta F_{\text{collapse}}$, a new surplus equilibrium exists at the post-upgrade level. Convergence to this equilibrium is basin-dependent; numerical verification confirms recovery from pre-shock equilibrium initial conditions across the parameter space studied (EC §EC.4.2).

(ii) Speed diagnostic (instantaneous-overload onset). *A secondary threshold*

$$\Delta F_{\text{rapid}} \;=\; \frac{S_0}{b\,F_0^{b-1}(1 + aT^*)}$$

marks the boundary above which the service gap turns positive at $t = 0^+$, initiating a rapid self-reinforcing collapse to the low-trust attractor. Because $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ at baseline (0.40 vs. 0.31), a firm upgrading by $\Delta F \in [\Delta F_{\text{collapse}}, \Delta F_{\text{rapid}})$ collapses permanently but through a slow *trajectory invisible on a standard fiscal-quarter horizon. The safety threshold is $\Delta F_{\text{collapse}}$; ΔF_{rapid} characterises collapse speed, not safety.*

Step 1: Two thresholds. The permanent-collapse threshold follows directly from Theorem 1: for $F_0 + \Delta F \geq F_{\text{crit}}$, the surplus residual $\Phi(S)$ (Lemma 2) at $S = 0$ gives $\Phi(0) \leq 0$, so no surplus equilibrium exists. Therefore

$$\Delta F_{\text{collapse}} = F_{\text{crit}} - F_0 = \Big(\frac{u}{\delta}\, h\Big(\frac{f}{f+\mu}\Big)\Big)^{1/b} - F_0.$$

At baseline ($F_0 = 0.80$): $\Delta F_{\text{collapse}} = 1.1104 - 0.80 = 0.310$.

For $\Delta F < \Delta F_{\text{collapse}}$, the post-upgrade $F_0 + \Delta F < F_{\text{crit}}$, so a new surplus equilibrium exists (Lemma 2). Convergence to this equilibrium is basin-dependent; Remark 4 establishes boundedness, and numerical verification confirms convergence from pre-shock equilibrium initial conditions across the parameter configurations studied.

Step 2: Instantaneous-overload diagnostic. The first-order demand increment after the upgrade is $\Delta D \approx (1 + aT^*)\, b\, F_0^{b-1}\, \Delta F$. Setting $\Delta D = S_0$ (the pre-upgrade surplus):

$$\Delta F_{\text{rapid}} = \frac{S_0}{b\, F_0^{b-1}(1 + aT^*)}. \tag{6}$$

At baseline: $\Delta F_{\text{rapid}} = 0.401$. Since $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ (0.401 vs. 0.310), the interval $[\Delta F_{\text{collapse}}, \Delta F_{\text{rapid}})$ defines a *slow-collapse zone*: the surplus equilibrium is gone, but $G(t_0^+) < 0$, so collapse proceeds through a gradual erosion of surplus rather than immediate overload. For $\Delta F \geq \Delta F_{\text{rapid}}$, $G(t_0^+) > 0$ and collapse is rapid.

Step 3: Dynamics and timescales. Under overload, the tipping threshold $W_{\text{tip}} = f/(f + \mu + \sigma G_0^+)$ marks where depletion exceeds recovery. Two timescales govern speed: $\tau_{\text{adapt}} = 1/r \approx 2.5$ mo and $\tau_{\text{deplete}} = 1/(\mu + \sigma G_0^+) \in [1.5, 4]$ mo. Near $\Delta F_{\text{collapse}}$, convergence may require hundreds of months.

Step 4: Three zones. $\Delta F_{\text{collapse}}$ and ΔF_{rapid} define: (i) *safe zone* ($\Delta F < \Delta F_{\text{collapse}}$), where a surplus equilibrium exists; (ii) *slow-collapse zone* ($\Delta F_{\text{collapse}} \leq \Delta F < \Delta F_{\text{rapid}}$), where $G(t_0^+) < 0$ but no surplus equilibrium exists, so collapse is gradual; (iii) *rapid-collapse zone* ($\Delta F \geq \Delta F_{\text{rapid}}$), where $G(t_0^+) > 0$ immediately. The ordering $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ reverses at high f (f=0.10: $\Delta F_{\text{collapse}} = 0.49 > \Delta F_{\text{rapid}} \approx 0.40$). Simulations ($t_{\text{end}}$=500, collapse criterion $T(\infty) < 0.15$) confirm collapse boundaries across three workforce scenarios at F_0=0.80. Minor discrepancies between simulated and analytic thresholds reflect slow-convergence transients near the boundary and basin-boundary effects. The correct safety threshold is $\Delta F_{\text{collapse}}$; ΔF_{rapid} characterises collapse speed, not safety. □

1.6. Proof of Proposition 4 (Complementarity)

PROPOSITION 4 **(Digitalisation-Workforce Complementarity, restated).** *$\partial K_{\text{eff}}/\partial W = pK > 0$ and $\partial^2 K_{\text{eff}}/\partial W \partial K = p > 0$. Workforce well-being determines $F_{\max}(W) = (K^* h(W))^{1/b}$, which is strictly increasing in W.*

Part 1: Cross-partial. From $K_{\text{eff}} = K \cdot [(1-p) + pW]$: $\partial K_{\text{eff}}/\partial W = pK$ and $\partial^2 K_{\text{eff}}/\partial W \partial K = p$. Both are strictly positive for $p > 0$.

Part 2: $F_{\max}(W)$ **at the regime boundary.** $F_{\max}(W)$ is defined as the value of F at which the surplus equilibrium of Lemma 2 ceases to exist, i.e., at which $S^* = 0$. From Lemma 2, at $S^* = 0$:

$$T^*(S) = \frac{\beta S}{\beta S + \lambda} \xrightarrow{S \to 0} 0.$$

Therefore, substituting $T^* = 0$ into the F_{crit} formula from Theorem 1 (Eq. 1 at $S^* = 0$):

$$\begin{aligned} F_{\max}(W) &= \Big(K^* \cdot [(1-p) + pW]\Big)^{1/b} \\ &= (K^*\, h(W))^{1/b}. \end{aligned}$$

This boundary formula is exact at $S^* = 0$ because $T^* \to 0$ is a direct consequence of the equilibrium structure, not an approximation.

Full implicit differentiation at interior equilibria ($S^* > 0$**).** At interior equilibria, $T^* > 0$ and $F_{\max}$ solves the fixed-point equation $(K^* h(W))/(1 + aT^*(W, F_{\max})) = F_{\max}^b$. Total differentiation with respect to W yields:

$$\frac{dF_{\max}}{dW} = \frac{1}{b} F_{\max}^{1-b} \frac{d}{dW}\left[\frac{K^* h(W)}{1 + aT^*}\right],$$

where

$$\frac{d}{dW}\left[\frac{K^* h(W)}{1 + aT^*}\right] = \frac{K^* p}{1 + aT^*} - \frac{K^* h(W)\, a}{(1 + aT^*)^2} \frac{dT^*}{dW}.$$

Since $T^*(S^*)$ is increasing in S^* and $dS^*/dW > 0$ (Theorem 2, Link 2–Link 3), the correction term $-K^* h(W) a\, (dT^*/dW)/(1 + aT^*)^2$ is negative, meaning the interior-equilibrium slope $dF_{\max}/dW$ is

Table 1 Sensitivity of F_{crit} to tanh smoothing bandwidth ε.

Smoothing ε	F_{crit}	Change vs. exact (%)
Exact ($\varepsilon = 0$)	1.110	–
$\varepsilon = 0.001$	1.106	-0.09
$\varepsilon = 0.005$	1.104	-0.27
$\varepsilon = 0.010$	1.100	-0.63
$\varepsilon = 0.050$	1.078	-2.62

The smoothing bandwidth ε replaces exact $[\cdot]^+$ operators with $\frac{1}{2}(x + \sqrt{x^2 + \varepsilon^2})$. All three theorems are qualitatively invariant for $\varepsilon \leq 0.01$.

somewhat smaller than the boundary-formula slope $(p/b)\,(K^*)^{1/b}[h(W)]^{1/b-1}$. However, the sign remains positive: $dF_{\max}/dW > 0$ for all $p > 0$, $K^* > 0$. The boundary formula therefore provides a conservative upper bound on the interior slope: the interior slope is smaller (by approximately 8% at baseline parameters), so the formula overstates how rapidly $F_{\max}$ increases with W at positive trust levels. $\square$

1.7. Proof of Corollary 5 (Budget-Constrained Trap)

COROLLARY 5 **(Budget-Constrained Trap, restated).** *Under a fixed budget $c_F F + c_f f \leq B$ with $B \geq B_{\min} - \varepsilon$ (where $B_{\min} = c_F \cdot F_{\mathrm{crit}}$), the CLV-maximising allocation includes $f^* > 0$: the all-F corner is strictly dominated.*

L et $B_{\min} \equiv c_F \cdot F_{\mathrm{crit}}$. *Case 1* ($B \geq B_{\min}$): The all-$F$ corner sets $F \geq F_{\mathrm{crit}}$, so Theorem 1 gives $\partial\,\mathrm{CLV}/\partial F \leq 0$ while Theorem 2 gives $\partial\,\mathrm{CLV}^*/\partial f > 0$; reallocating the marginal dollar to f yields a strict CLV gain. *Case 2* ($B < B_{\min}$): A surplus equilibrium exists. Theorem 2 guarantees $\partial\,\mathrm{CLV}^*/\partial f > 0$, so the all-$F$ corner forgoes a positive return. Near $B_{\min}$, per-dollar returns are continuous (IFT on $W^*(S), T^*(S)$), and Case 1's strict dominance persists in a neighbourhood $B \in (B_{\min} - \varepsilon, B_{\min})$. For $B \in [B_{\min}/2, B_{\min} - \varepsilon)$, dominance holds computationally across $c_F/c_f \in [0.5, 5]$ at baseline values of all other parameters. $\square$

1.8. Smoothing Robustness

Table 1 verifies that replacing the exact piecewise-smooth operators $[\cdot]^+$ with a tanh-smoothed approximation shifts F_{crit} by less than 1% for $\varepsilon \leq 0.01$, confirming that all three theorems are invariant to smoothing.

Having established that the analytical results are robust to smoothing, we now show that the model nests the foundational quality-erosion framework as a special case.

Table 2 Structural mapping between WCT (restricted) and Oliva–Sterman (2001).

WCT term	O&S counterpart	Correspondence
$r\,[G]^+\,W$	Learning from failures	Product of gap and workforce effort
δK	Depreciation	Constant-rate technology decay
$([-G]^+ + f)(1-W)$	Recovery + staffing	Surplus and investment raise effort; saturates at $W{=}1$
$(\mu + \sigma\,[G]^+)\,W$	Corner-cutting + fatigue	Overload and decay deplete effort; vanishes at $W{=}0$
Trust stock T	No O&S counterpart	Collapses to demand at $\beta \to \infty$, $\lambda = 0$

Under the three restrictions (p=0, a=0, F constant), the WCT trust stock collapses into the demand function, recovering the O&S perceived-quality attractiveness mechanism. The $(1-W)$ saturation factor replaces O-S's nonlinear lookup table.

2. Structural Mapping: WCT and Oliva–Sterman (2001)

The WCT model recovers the core dynamics of Oliva and Sterman (2001) as a special case under three parameter restrictions: (i) $p = 0$ (workforce does not affect K_{eff}, so $h(W) = 1$); (ii) $a = 0$ (no trust-demand feedback, so $D = F^b$); and (iii) F constant (no cross-domain dynamics). Under these settings, the trust ODE remains active but effectively decouples from demand; setting additionally $\beta \to \infty$ and $\lambda = 0$ collapses T into the demand function, recovering the O-S perceived-quality attractiveness mechanism. Table 2 provides the term-by-term correspondence.

Qualitative behaviour is identical. With the three restrictions lifted, the model generates Loops R1, R2, and the cross-domain Front-Stage Trap.

3. Sensitivity Analysis

Table 3 reports how each theorem's qualitative result responds to $\pm 50\%$ perturbations of each parameter from its baseline value.

Key observations. p is the most uniformly influential parameter: reducing it weakens all three theorems simultaneously. b is critical for Theorem 1 (at $b = 1$ the Trap disappears); σ for Theorem 3 (at $\sigma = 1$ bistability vanishes). No single perturbation kills all three results.

CLV calibration. The customer-lifetime-value mapping $\text{CLV}(T) = w(T)/(1 - \rho(T))$ uses illustrative parameters ($\rho_0 = 0.80$, $w_0 = 80$). Because both $\rho(\cdot)$ and $w(\cdot)$ are monotonically increasing in T, and T^* itself is monotone in f (Theorem 2), all qualitative predictions (sign of $\partial\,\text{CLV}^*/\partial f$, existence of the Cliff, complementarity) are invariant to the CLV parametrisation. Only quantitative magnitudes (e.g., the 40% CLV gap between scenarios) depend on these choices.

Table 3 Sensitivity of theorem predictions to $\pm 50\%$ parameter perturbations. †Slower sensing ($r\downarrow$) allows overload more time to erode W before the self-correction loop activates, steepening the cliff.

Parameter perturbation	Thm 1 (Trap)	Thm 2 (Buffer)	Thm 3 (Cliff)
b: 1.3 $\to$ 1.0	Disappears	Holds	Weakens
b: 1.3 $\to$ 1.6	Strengthens	Holds	Strengthens
a: 0.4 $\to$ 0.2	Weakens	Holds	Holds
a: 0.4 $\to$ 0.6	Strengthens	Holds	Holds
p: 0.5 $\to$ 0.25	Weakens	Weakens	Weakens
p: 0.5 $\to$ 0.75	Strengthens	Strengthens	Strengthens
σ: 2.0 $\to$ 1.0	Weakens	Holds	Weakens
σ: 2.0 $\to$ 3.0	Strengthens	Holds	Strengthens
η: 2.0 $\to$ 1.0	Weakens	Holds	Holds
η: 2.0 $\to$ 3.0	Strengthens	Holds	Holds
r: 0.4 $\to$ 0.2	Holds	Holds	Strengthens†
r: 0.4 $\to$ 0.8	Holds	Holds	Weakens
δ: 0.03 $\to$ 0.015	Weakens	Strengthens	Weakens
δ: 0.03 $\to$ 0.06	Strengthens	Weakens	Strengthens

Each cell reports the qualitative effect of the perturbation on the theorem's central prediction. *Holds*: qualitative sign unchanged relative to baseline. *Weakens*: effect direction preserved but threshold shifts toward baseline (smaller safe zone or weaker buffer). *Strengthens*: effect becomes more pronounced. *Disappears*: result no longer obtains at the perturbation value. All perturbations are $\pm 50\%$ from the baseline in Table 2 of the main paper.

Table 4 Supplementary sensitivity: leading-indicator lag and direct-channel share across key parameter ranges.

Outcome	Parameter varied	Low	Baseline	High
Leading-indicator lag (months)†	$\beta \in \{0.06, 0.12, 0.24\}$, $\sigma = 2.0$	21	12	6
	$\sigma \in \{1.0, 2.0, 3.0\}$, $\beta = 0.12$	11	12	12
Indirect-channel share (%)	$p \in \{0.25, 0.50, 0.75\}$	<1	<1	<1

†Leading-indicator lag: months by which W leads T before T crosses the 0.10 crisis threshold. Scenario: step upgrade $\Delta F = 0.40$ from $F_0 = 0.80$ (above $\Delta F_{\text{collapse}} = 0.31$), no workforce investment, starting from the surplus equilibrium. Lower β (stickier customer trust) lengthens the warning window; σ variation contributes less than two months at fixed β. Indirect-channel share: fraction of CLV uplift from f via the sensing channel (p=0 counterfactual) at $F = 0.85$; values $< 1\%$ confirm the capacity-multiplier channel dominates. See §4.3 of the main paper.

4. Calibration Protocol

4.1. Parameter Sources

All time units are months; rate parameters are in months^{-1} with half-lives $\ln(2)/\theta$. The calibration follows Homer (1993) and Barlas (1996); baseline values, timescales, kill conditions, and empirical anchors are reported in Table 2 of the main paper. Three parameters warrant additional discussion.

Demand super-linearity ($b = 1.3$). A 10% front-stage increase raises back-stage demand by approximately 13%. The Trap requires only $b > 1$; sensitivity in Table 3.

Table 5 Simulation initial conditions and controls by figure.

Fig.	(K_0, W_0, T_0)	F_0	u	f	$t_{\max}$	Notes
2	balance†	varies	0.05	0.03	500	37×26 grid; CLV thr. $= 600$
3	Analytical: $F_{\max}(W) = (K^* h(W))^{1/b}$; $K^* = u/\delta$					Boundary formula
4	$(1.60, 0.55, 0.50)$	0.80	0.05	varies	120	Three scenarios (A,B,C)
5	(K^*, W^*, T^*)	0.80	0.05‡	0.01–0.10	500	61-pt ΔF sweep
EC	eq. IC at F=0.85	0.85	0.05	0.00–0.20	120	Study 3: Thm 2 decomp.; p=0 counterfactual confirms structural-zero

†Balance manifold: $K_0 h(W_0) = (1+a\cdot 0.5)F^b$, T_0=0.50; collapse criterion: $T(500)<0.15$.
‡Inherited from equilibrium.

Workforce-technology complementarity ($p = 0.5$). A burned-out workforce ($W = 0$) delivers 50% of design capacity; the value is illustrative and all results hold for any $p \in (0, 1)$.

Overload-recovery asymmetry ($\sigma = 2.0$). Magnitude comparator from Oliva and Sterman (2001); independently consistent with the loss-aversion ratio $\lambda \approx 2.25$ from prospect theory (Tversky and Kahneman 1992). The Cliff disappears only at $\sigma = 1$ (Table 3).

4.2. Simulation Initial Conditions

Table 5 reports initial conditions for each figure. All simulations use baseline parameters from Table 2 of the main paper, starting in the surplus regime ($K_{\text{eff}} > D$). Basin-of-attraction grid: 37×26 trajectories; doubled-grid validation (73×51) shifts the separatrix by at most 0.0075 W_0-units.

4.3. CLV Functional Form

We model CLV using the geometric-series structure of Gupta et al. (2004): CLV $= w(T)/(1-\rho(T))$, where $\rho(T) = 0.80 + 0.18T$ is an annual retention probability and $w(T) = 80(1 + 0.25\max(T - 0.15, 0))$ an annualised per-customer margin. The retention endpoints ($\rho(0) = 0.80$, $\rho(1) = 0.98$) match the 70–90% annual range in Gupta et al. (2004); \$80 is an illustrative annual margin for US omnichannel grocery. The model's analytical results require only $\partial\,\text{CLV}/\partial T^* > 0$, which holds everywhere since $\partial\,\text{CLV}/\partial T^* \geq w(T^*)\rho'(T^*)/(1-\rho(T^*))^2 > 0$.

5. Empirical Appendix

This appendix supplements the empirical analysis in §5 of the main paper.

US Data and Robustness

US data sources. The US firm-level panel assembles four longitudinal data sources. Customer satisfaction is measured by the American Customer Satisfaction Index (ACSI; scores 0–100 sourced from `theacsi.org`). Workforce well-being is proxied by Glassdoor overall ratings (1–5 scale), restricted to firm–years with at least five employee reviews to ensure representativeness. Financial data, including capital expenditure and revenue, are drawn from SEC EDGAR 10-K filings accessed via the XBRL API. National e-commerce penetration uses the Federal Reserve FRED series ECOMPCTSA; labour-market controls use BLS JOLTS vacancy rates. The empirical proxies map to model variables as follows: $K =$ CapEx/Revenue (technology stock), $W =$ Glassdoor overall rating (workforce well-being), $T =$ ACSI score (customer trust), and the interaction $K^z \times W^z$ uses standardised values.

US panel construction. The panel is constructed using EDGAR financial data as the backbone (276 firm–year observations across 17 firms, 2008–2024). ACSI, Glassdoor, and macro controls are then left-joined to this backbone using firm–year keys. All merges enforce one-to-one key uniqueness; no duplicate keys were detected. The resulting panel contains 131 complete firm–year observations with non-missing values for all four key variables, representing 47% coverage of the full backbone. The complementarity specification therefore uses 131 observations across 13 firms; the inverted-U and event-study specifications, which require only ACSI and e-commerce share, use 216 and 108 observations across 17 and 16 firms, respectively.

US robustness checks. Six robustness checks supplement the core between-effects OLS estimand (reported in §5.2 of the main paper): (i) Mundlak CRE, (ii) one-year lag ($K^z_{t-1} \times W^z_{t-1}$), (iii) COVID exclusion (dropping 2020–2021), (iv) leave-one-out, (v) wild cluster bootstrap (9,999 reps, Webb six-point weights; Webb 2023), and (vi) CR3 small-sample correction ($\mathrm{df} = G - k = 9$). All checks preserve the positive sign.

Event-Study Pre-Trends

Table 6 reports event-time coefficients (§5.2; $N = 108$, 16 firms). Pre-period coefficients are insignificant in both specifications, supporting parallel trends.

Event-Study Launch Identification

The event study identifies the launch year as the fiscal year in which each firm's primary omnichannel fulfilment service (curbside pickup, click-and-collect, or same-day delivery) reached nationwide or near-nationwide availability. Launch years are identified from company press releases, investor presentations, and 10-K disclosures. Representative examples: Kroger ClickList (2016), Walmart Grocery Pickup (2017), Target Drive Up (2018), Albertsons Drive Up & Go

Table 6 **Event-time coefficients: parallel-trends check.**

	Level DID			*Moderation* ($\times W^z_{\text{pre}}$)		
Event time	Coef.	t-stat	p-value	Coef.	t-stat	p-value
$t=-3$	-0.125	-0.40	0.696	-2.093	-0.11	0.918
$t=-2$	-0.375	-1.15	0.269	-2.174	-0.11	0.914
$t=-1$		(reference)			(reference)	
$t=0$	-0.367	-2.21	0.043	-1.266	-0.06	0.952
$t=+1$	-0.167	-0.65	0.528	-0.876	-0.04	0.967
$t=+2$	-0.300	-0.78	0.447	-0.875	-0.04	0.967
$t=+3$	-0.900	-1.69	0.113	-1.033	-0.05	0.961

Dependent variable: ACSI score. Firm FE; cluster-robust SEs with p-values from $t(15)$. Reference period: $t=-1$. Level DID: event-time dummies only. Moderation: interaction of event-time dummies with pre-shock standardised Glassdoor (W^z_{pre}). All pre-period coefficients ($t=-3$, $t=-2$) are insignificant in both specifications.

(2017), Costco same-day (2017). Amazon is excluded because its omnichannel transition was gradual. The complete 16-firm launch mapping is documented in the replication code (`09_event_study.py`).

EU Replication (Exploratory)

An independent panel of 11 European retailers (UK, France, Germany, Spain, Sweden, Netherlands; 2019–2023; N=20) uses Trustpilot for satisfaction, Glassdoor for well-being, and Yahoo Finance for financials (asset percentile rank as currency-neutral control). E-commerce share from UK ONS and Eurostat.

The between-effects interaction is positive (+0.47; CR2: $p = 0.045$) but fails robustness (HC3: $p = 0.53$; Webb bootstrap: $p = 0.62$), reflecting $N_c = 11$ clusters (Cameron et al. 2008, Webb 2023). Additional checks (currency-neutral controls, UK-only/Continental splits, leave-one-out with $K\times W \in [-0.04, +0.82]$, Imbens–Kolesar df_{eff}=7.2, p=0.055) are reported as exploratory.

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